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Bipartite and $\mathcal{H}_\infty$ bipartite synchronization for multiweighted coupled fractional-order delayed reaction-diffusion neural networks

Ya-Nan Li [a], Jin-Liang Wang [a,b,*], Shun-Yan Ren [c], Tingwen Huang [d]

[a] Tianjin Key Laboratory of Autonomous Intelligence Technology and Systems, School of Computer Science and Technology, Tiangong University, Tianjin, 300387, China
[b] School of Information Science and Engineering, Linyi University, Linyi, 276005, China
[c] School of Intelligent Manufacturing and Electrical Engineering, Guangzhou Institute of Science and Technology, Guangzhou, 510540, China
[d] Faculty of Computer Science and Control Engineering, Shenzhen University of Advanced Technology, Shenzhen, 518055, China

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ABSTRACT

In this paper, two types of multiweighted coupled fractional-order delayed reaction-diffusion neural networks (MCFDRNN) with and without external disturbances under signed graph are introduced, which generalize the existing coupled fractional-order reaction-diffusion networks models. Several bipartite and $\mathcal{H}_\infty$ bipartite synchronization conditions for these two MCFDRNN are given on the basis of some important lemmas and matrix theory, which can not be acquired by utilizing the method and technique used in coupled reaction-diffusion neural networks. Finally, the derived criteria are validated by exploiting a numerical example.

1. Introduction

Since successful applications in numerous fields (e.g., image encryption Hui et al. (2024), secure communication Wu et al. (2023b), image recognition Mijwel et al. (2023), etc.), the synchronization in coupled reaction-diffusion neural networks (CRNN) has been deeply discussed and considerable valuable results have been acquired (Guo et al., 2005; Wang et al., 2022, 2023). In Wang et al. (2023), a pinning sampled data control scheme was developed to address the synchronization for a CRNN with deception attacks and spatiotemporal sampled-data coupling. Guo et al. (2005) utilized pinning control approach to investigate the global exponential synchronization for a coupled delayed memristive reaction-diffusion neural networks. In Wang et al. (2022), a fuzzy CRNN with external disturbances was introduced, and several $\mathcal{H}_\infty$ synchronization criteria for such network were derived on the basis of an adaptive pinning control strategy. However, in the above introduced works (Guo et al., 2005; Wang et al., 2022, 2023), they are confined to single weighted network models.

Due to the wide range of influencing factors, it is very essential and significant to further discuss multiweighted CRNN. Up to now, plenty of meaningful results about synchronization for multiweighted CRNN have been acquired (Cao et al., 2022; Lin & Liu, 2023; Zhao et al., 2023). In Lin and Liu (2023), several synchronization criteria were derived for a multiweighted CRNN with state and spatial diffusion couplings by employing pinning control method. Zhao et al. (2023) utilized pinning and adaptive control methods to deal with the robust $\mathcal{H}_\infty$ synchronization for two types of CRNN, in which the nodes are coupled through their states in the first one and the nodes are coupled through spatial diffusion terms in the second one. In Cao et al. (2022), by employing a pinning impulsive control strategy, several synchronization conditions were given for a multiweighted coupled stochastic reaction-diffusion neural networks.

Nevertheless, these existing works on synchronization are all focused on the integer-order network models for not only the single weighted CRNN (Guo et al., 2005; Wang et al., 2022, 2023) but also the multiple weighted CRNN Cao et al. (2022), Lin and Liu (2023), Zhao et al. (2023).

In fact, due to the characteristics of heredity and memory (Asker et al., 2025; Damak, 2023), fractional-order derivative can more accurately model the dynamics of CRNN compared with integer-order derivative. In view of this fact, some authors have further tackled the synchronization for coupled fractional-order reaction-diffusion neural networks (CFRNN) (Bao & Li, 2025; Lv et al., 2020; Wang et al., 2024; You et al., 2024). In Lv et al. (2020), several adaptive control strategies to tune all or a part of the coupling weights were developed for ensuring the synchronization of CFRNN with time-dependent or space-time-dependent coupling weights. Bao and Li Bao and Li (2025) introduced a CFRNN with output coupling, and utilized the event-triggered control method to obtain an output synchronization criterion for such network. Wang et al. (2024) derived several synchronization criteria for a multiweighted CFRNN by selecting appropriate integer-order or

* Corresponding author.
E-mail addresses: 19065460385@163.com (Y.-N. Li), wangjinliang1984@163.com (J.-L. Wang), renshunyan@163.com (S.-Y. Ren), huangtingwen2013@gmail.com (T. Huang).

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Table 1
The comparison between existing results and our work.

Characteristics	Multiple weights	Fractional-order	Bipartite synchronization	$\mathcal{H}_\infty$ Bipartite synchronization
Guo et al. (2005), Wang et al. (2022, 2023), Ahmed et al. (2023), Ali et al. (2025a,b, 2022, 2017), Alsarhan et al. (2024), Ayad (2024), Barazanchi et al. (2024), Rehman et al. (2025), Samy et al. (2025), Tian et al. (2025)	×	×	×	×
Cao et al. (2022), Lin and Liu (2023), Zhao et al. (2023)	✓	×	×	×
Bao and Li (2025), Lv et al. (2020),Sharafian et al. (2025, 2024)	×	✓	×	×
Wang et al. (2024), You et al. (2024)	✓	✓	×	×
Hao et al. (2024), Wu et al. (2023a), Xu et al. (2024)	×	✓	✓	×
Ours	✓	✓	✓	✓

fractional-order coupling weight updating strategies and utilizing output strict passivity. In You et al. (2024), two types of coupled fractional-order memristive reaction-diffusion neural networks with multiple output or output derivative couplings were presented, and several output synchronization criteria were given for these two networks based on the devised adaptive control schemes.

It should be pointed out that the coupling weights among nodes are required to be non-negative in these existing results (Asker et al., 2025; Bao & Li, 2025; Cao et al., 2022; Damak, 2023; Guo et al., 2005; Lin & Liu, 2023; Lv et al., 2020; Wang et al., 2022, 2024, 2023; You et al., 2024; Zhang et al., 2025; Zhao et al., 2023, 2025), In fact, the interactions among nodes in some real networks (e.g., social networks, economic systems, and so on Hao et al. (2024)) may be competitive besides cooperative. Apparently, it may be better to describe the competition-cooperation interactions among nodes by utilizing the sign graph. Compared with the synchronization in CFRNN with unsigned graph, CFRNN with sign graph can exhibit the more meaningful bipartite synchronization and several valuable results in this direction have been obtained (Hao et al., 2024; Wu et al., 2023a; Xu et al., 2024). In Hao et al. (2024), a bipartite synchronization criterion for a coupled delayed fractional-order neural networks with reaction-diffusion terms were derived on the basis of M-matrix theory. Xu et al. (2024) utilized fuzzy-based aperiodically intermittent control approach to address the bipartite synchronization for a coupled fractional-order heterogeneous reaction-diffusion neural networks. In Wu et al. (2023a), gave some sufficient conditions to ensure the bipartite synchronization in a CFRNN with time-varying delay by designing a quantized pinning state feedback control scheme. Table 1 provides a comparison between the existing results and our work.

The above-mentioned works focused mainly on bipartite synchronization of single weighted CFRNN without external disturbances (Hao et al., 2024; Wu et al., 2023a; Xu et al., 2024). Regrettably, the bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization for multi-weighted coupled fractional-order delayed reaction-diffusion neural networks (MCFDRNN) has not yet been discussed until now. This paper respectively tackles the bipartite synchronization for MCFDRNN with and without external disturbances. The three main contributions of this paper are listed as follows.

1. Two types of MCFDRNN with signed graph are introduced, which not only generalize some existing multiweighted CRNN models but also can better reflect the interaction relationships among fractional-order delayed reaction-diffusion neural networks.
2. A bipartite synchronization criterion for the MCFDRNN is given by employing the matrix theory and the fractional-order differential inequality, which lays a theoretical foundation to further deal with the bipartite synchronization in MCFDRNN.
3. The $\mathcal{H}_\infty$ bipartite synchronization for the MCFDRNN with external disturbances and signed graph is also tackled, which not only clearly reflects the influences of external disturbances on the bipartite synchronization but also is completely different from the bipartite synchronization in their proof methods.

2. Preliminaries

Several useful notations, essential graph theory, basic definition, and necessary lemmas are introduced in this section.

2.1. Notations

Let $\mathbb{R}^+ = [0, +\infty)$. For a symmetric matrix $A_1 \in \mathbb{R}^{q_1\times q_1}$, $\xi_1(A_1), \xi_2(A_1), \ldots,$ and $\xi_{q_1}(A_1)$ denote the eigenvalues of A_1 and satisfy $\xi_1(A_1) \geqslant \xi_2(A_1) \ldots \geqslant \xi_{q_1}(A_1)$. $\mathbb{R}^{q_2\times q_2} \ni A_2 > (\geqslant, \leqslant)0$ indicates that A_2 is a positive (semi-positive, semi-negative) definite and symmetric matrix. $\mathcal{L}\{\cdot\}$ is the Laplace transform. $\Theta = \{\theta = (\theta_1, \theta_2, \ldots, \theta_\upsilon)^T \in \mathbb{R}^\upsilon | \quad |\theta_a| < g_a, 0 < g_a \in \mathbb{R}, a = 1, 2, \ldots, \upsilon\}$. $\partial\Theta$ denotes the boundary of Θ and $\bar{\Theta} = \Theta \cup \partial\Theta$. For any $\phi(\theta, t) = (\phi_1(\theta, t), \phi_2(\theta, t), \ldots, \phi_{q_3}(\theta, t))^T \in \mathbb{R}^{q_3}$ with $(\theta, t) \in \Theta \times \mathbb{R}$, we define

$$\|\phi(\cdot, t)\| = \sqrt{\sum_{i=1}^{q_3} \int_\Theta \phi_i^2(\theta, t) d\theta}.$$

2.2. Graph theory

Let $\mathcal{V} = \{1, 2, \ldots, \zeta\}$ and $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ be the node set and undirected edge set respectively. In this paper, $\mathcal{V}$ can be divided into two nonempty subsets $\mathcal{V}_1$ and $\mathcal{V}_2$, in which $\mathcal{V}_1 \cup \mathcal{V}_2 = \mathcal{V}$ and $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$. $Q = (Q_{\epsilon\omega})_{\zeta\times\zeta} \in \mathbb{R}^{\zeta\times\zeta}$ indicates the weighted adjacency matrix, in which $Q_{\epsilon\omega} \in \mathbb{R}$ is denoted below:

$$Q_{\epsilon\omega} = \begin{cases} Q_{\omega\epsilon} > 0, & (\epsilon, \omega) \in \mathcal{E},\ \epsilon, \omega \text{ stay the identical set } \mathcal{V}_u, \\ Q_{\omega\epsilon} < 0, & (\epsilon, \omega) \in \mathcal{E}, \text{ but } \epsilon, \omega \text{ stay the different sets } \mathcal{V}_1 \text{ and } \mathcal{V}_2, \\ -\sum\limits_{\substack{\beta=1 \\ \beta\neq\epsilon}}^{\zeta} |Q_{\epsilon\beta}|, & \omega = \epsilon, \\ 0, & \text{otherwise}, \end{cases}$$

where $u \in \{1, 2\}$, and $Q_{\epsilon\omega} > 0 (< 0)$ represents cooperative(competitive) relationship. Define $s_\epsilon = 1$ if $\epsilon \in \mathcal{V}_1$ and $s_\epsilon = -1$ if $\epsilon \in \mathcal{V}_2$. Since

$$\begin{aligned} & s_1 Q_{1j} + s_2 Q_{2j} + \ldots + s_j Q_{jj} + \ldots + s_\zeta Q_{\zeta j} \\ &= s_j \left(s_1 s_j Q_{1j} + s_2 s_j Q_{2j} + \ldots + s_j s_j Q_{jj} + \ldots + s_\zeta s_j Q_{\zeta j} \right) \\ &= s_j \left[|Q_{1j}| + |Q_{2j}| + \ldots + \left(-\sum_{\substack{\beta=1 \\ \beta\neq j}}^{\zeta} |Q_{\beta j}| \right) + \ldots + |Q_{\zeta j}| \right] \\ &= 0,\ j = 1, 2, \ldots, \zeta, \end{aligned}$$

one has $s^T Q = 0$, in which $s = (s_1, s_2, \ldots, s_\zeta)^T$. Based on this fact, it is not difficult to derive that the sum of each row is zero and off-diagonal elements are nonnegative for SQS, in which $S = \mathrm{diag}(s_1, s_2, \ldots, s_\zeta) \in \mathbb{R}^{\zeta\times\zeta}$.

2.3. Definition

Definition 1. (see Podlubny, 1999): The Caputo fractional-order derivative for continuously differentiable function $f(t)$ is given as follows:

$$D^{\varpi} f(t) = \frac{1}{\Gamma(1-\varpi)} \int_0^t \frac{\dot{f}(k)}{(t-k)^{\varpi}} dk,$$

in which $0 < \varpi < 1$.

2.4. Lemmas

Lemma 1. *(see Aguila-Camacho et al., 2014): Letting $\eta(t) \in C^1([0,+\infty), \mathbb{R}^b)$, we can acquire*

$$D^{\varpi}(\eta^T(t)K\eta(t)) \leqslant 2\eta^T(t)KD^{\varpi}\eta(t),$$

in which $\varpi \in (0,1)$ and $0 \leqslant K \in \mathbb{R}^{b\times b}$.

Lemma 2. *(see Lu, 2008): Let $\varsigma(\theta) \in C^1(\Theta, \mathbb{R})$ and $\varsigma(\theta)\mid_{\partial\Theta} = 0$. Then, one has*

$$\int_{\Theta} \varsigma^2(\theta)d\theta \leqslant g_a^2 \int_{\Theta} \left(\frac{\partial \varsigma(\theta)}{\partial \theta_a}\right)^2 d\theta, \ a = 1, 2, \dots, \upsilon.$$

Lemma 3. *(see He et al., 2018): If the following inequality holds:*

$$D^{\varpi} V(t) \leqslant -m_1 V(t) + m_2 \sup_{-\tau(t)\leqslant\varepsilon\leqslant 0} V(t+\varepsilon),$$

one then gets

$$\lim_{t\to+\infty} V(t) = 0,$$

in which $0 < \varpi < 1$, $0 < m_2 < m_1 \in \mathbb{R}$, $0 < \psi \in \mathbb{R}$, $V(t) \in C([-\psi,+\infty), \mathbb{R}^+)$, and $\tau(t) \in C(\mathbb{R}^+, \mathbb{R}^+)$ satisfies the following requirements:

$$\tau(t) \leqslant t + \psi \text{ and } \lim_{t\to+\infty}(t-\tau(t)) = +\infty.$$

Lemma 4. *(see Aguila-Camacho et al., 2014): For an irreducible matrix $M = (M_{\epsilon\omega})_{\zeta\times\zeta} \in \mathbb{R}^{\zeta\times\zeta}$, we have*

$$\ell^T(M \otimes N)\ell \leqslant \xi_2(M)\ell^T(I_{\zeta} \otimes N)\ell,$$

in which $\ell = (\ell_1^T, \ell_2^T, \dots, \ell_{\zeta}^T)^T \in \mathbb{R}^{\epsilon v}$ with $\sum_{q=1}^{\zeta} \ell_q = 0$, $0 < N \in \mathbb{R}^{v\times v}$, and $M_{\epsilon\omega} \in \mathbb{R}$ satisfies

$$M_{\epsilon\omega} = \begin{cases} M_{\omega\epsilon} \geqslant 0, & \text{if } \omega \neq \epsilon, \\ -\sum_{\substack{q=1\\ q\neq\epsilon}}^{\zeta} M_{\epsilon q}, & \text{if } \omega = \epsilon. \end{cases}$$

3. Bipartite synchronization and $\mathcal{H}_{\infty}$ bipartite synchronization for MCFDRNN

In this section, a bipartite synchronization condition for the CFDRNN is given on the basis of the Caputo fractional-order derivative properties and inequality techniques.

3.1. CFDRNN

A type of MCFDRNN with signed graph is characterized by the following fractional-order delayed partial differential equations:

$$\begin{aligned} D^{\varpi}\kappa_{\epsilon}(\theta,t) &= W\Delta\kappa_{\epsilon}(\theta,t) - H\kappa_{\epsilon}(\theta,t) + B\hbar(\kappa_{\epsilon}(\theta,t)) \\ &\quad + \sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta} d^v Q_{\epsilon\omega}^v P^v \kappa_{\omega}(\theta,t) \\ &\quad + B\hbar(\kappa_{\epsilon}(\theta,t-\tau(t))), \ \epsilon = 1,2,\dots,\zeta, \\ \kappa_{\epsilon}(\theta,t) &= 0, (\theta,t) \in \partial\Theta \times [-\psi,+\infty), \\ \kappa_{\epsilon}(\theta,t) &= \Psi_{\epsilon}(\theta,t), (\theta,t) \in \Theta \times [-\psi,0], \end{aligned} \tag{1}$$

in which $\varpi \in (0,1)$; $\kappa_{\epsilon}(\theta,t) = (\kappa_{\epsilon 1}(\theta,t), \kappa_{\epsilon 2}(\theta,t), \dots, \kappa_{\epsilon\iota}(\theta,t))^T \in \mathbb{R}^{\iota}$ is state vector of the node ϵ; $\Delta = \sum_{a=1}^{\upsilon} \frac{\partial^2}{\partial\theta_a^2}$; $0 < W = \mathrm{diag}(w_1, w_2, \dots, w_{\iota}) \in \mathbb{R}^{\iota\times\iota}$, $0 < H = \mathrm{diag}(h_1, h_2, \dots, h_{\iota}) \in \mathbb{R}^{\iota\times\iota}$, $\hbar(\kappa_{\epsilon}(\theta,t)) = (\hbar_1(\kappa_{\epsilon 1}(\theta,t)), \hbar_2(\kappa_{\epsilon 2}(\theta,t)), \dots, \hbar_{\iota}(\kappa_{\epsilon\iota}(\theta,t)))^T \in \mathbb{R}^{\iota}$, and $B \in \mathbb{R}^{\iota\times\iota}$; $\Psi_{\epsilon}(\theta,t) \in \mathbb{R}^{\iota}$ is continuous on $\bar{\Theta} \times [-\psi,0]$; $\tau(t) \in \mathbb{R}$ is the time-varying delay, which satisfies $\lim_{t\to+\infty}(t-\tau(t)) = +\infty$ and $\tau(t) \leqslant t+\psi$ with $0 < \psi \in \mathbb{R}$; $0 < P^v = \mathrm{diag}(p_1^v, p_2^v, \dots, p_{\iota}^v) \in \mathbb{R}^{\iota\times\iota}$, $0 < d^v \in \mathbb{R}$, and $Q^v = (Q_{\epsilon\omega}^v)_{\zeta\times\zeta} \in \mathbb{R}^{\zeta\times\zeta}$ respectively denote the inner coupling matrix, the coupling strength, and the outer coupling matrix in the vth coupling form, in which $Q_{\epsilon\omega}^v \in \mathbb{R}$ has the same definition as $Q_{\epsilon\omega}$ in Section II.

In this section, topology structure for different coupling forms is identical in the network (1) and is connected. In addition, $\hbar_c(\cdot)$ meets

$$|\hbar_c(\delta_1) - \varrho\hbar_c(\delta_2)| \leqslant \vartheta_c|\delta_1 - \varrho\delta_2| \tag{2}$$

for any $\delta_1, \delta_2 \in \mathbb{R}$, in which $c = 1,2,\dots,\iota$, $\varrho \in \{1,-1\}$, and $\mathbb{R} \ni \vartheta_c > 0$. Define $\vartheta = \mathrm{diag}(\vartheta_1^2, \vartheta_2^2, \dots, \vartheta_{\iota}^2) \in \mathbb{R}^{\iota\times\iota}$.

A concept about the bipartite synchronization for the network (1) is given as follows.

Definition 2. The network (1) is said to reach the bipartite synchronization if

$$\lim_{t\to+\infty}\left\| s_{\epsilon}\kappa_{\epsilon}(\cdot,t) - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q\kappa_q(\cdot,t)\right\| = 0, \ \epsilon = 1,2,\dots,\zeta,$$

in which $s_{\epsilon} = 1$ if $\epsilon \in \mathcal{V}_1$ and $s_{\epsilon} = -1$ if $\epsilon \in \mathcal{V}_2$.

3.2. Bipartite synchronization for MCFDRNN

Taking $\bar{\kappa}(\theta,t) = \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q\kappa_q(\theta,t)$, one derives from (1) that

$$\begin{aligned} D^{\varpi}\bar{\kappa}(\theta,t) &= W\Delta\bar{\kappa}(\theta,t) - H\bar{\kappa}(\theta,t) + \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t)) \\ &\quad + \frac{1}{\zeta}\sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta} d^v\left(\sum_{q=1}^{\zeta} s_q Q_{q\omega}^v\right)P^v\kappa_{\omega}(\theta,t) \\ &\quad + \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t-\tau(t))) \\ &= W\Delta\bar{\kappa}(\theta,t) - H\bar{\kappa}(\theta,t) + \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t)) \\ &\quad + \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t-\tau(t))), \ \epsilon = 1,2,\dots,\zeta. \end{aligned}$$

Letting $\varphi_{\epsilon}(\theta,t) = (\varphi_{\epsilon 1}(\theta,t), \varphi_{\epsilon 2}(\theta,t), \dots, \varphi_{\epsilon\iota}(\theta,t))^T = s_{\epsilon}\kappa_{\epsilon}(\theta,t) - \bar{\kappa}(\theta,t)$, we have

$$\begin{aligned} D^{\varpi}\varphi_{\epsilon}(\theta,t) &= W\Delta\varphi_{\epsilon}(\theta,t) - H\varphi_{\epsilon}(\theta,t) + s_{\epsilon}B\hbar(\kappa_{\epsilon}(\theta,t)) \\ &\quad + s_{\epsilon}B\hbar(\kappa_{\epsilon}(\theta,t-\tau(t))) - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t)) \\ &\quad + \sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta} d^v s_{\epsilon}s_{\omega}Q_{\epsilon\omega}^v P^v\varphi_{\omega}(\theta,t) \\ &\quad - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t-\tau(t))), \end{aligned} \tag{3}$$

in which $\epsilon = 1,2,\dots,\zeta$.

In what follows, a bipartite synchronization criterion for the network (1) is provided.

Theorem 1. *If there are* $0 < E = (e_{cr})_{\iota\times\iota} \in \mathbb{R}^{\iota\times\iota}$ *and two positive constants* m_1, m_2 *such that*

$$EW + WE \geqslant 0, \tag{4}$$

$$m_1 > m_2 \geqslant \sqrt{\frac{\xi_1(\vartheta)}{\xi_\iota(E)}}, \tag{5}$$

$$\Xi + m_1 I_\zeta \otimes E + \sum_{v=1}^{\sigma} d^v (SQ^v S) \otimes (EP^v + P^v E) \leqslant 0, \tag{6}$$

in which $S = \mathrm{diag}(s_1, s_2, \ldots, s_\zeta) \in \mathbb{R}^{\zeta\times\zeta}$ *and* $\Xi = I_\zeta \otimes \Big[-\sum_{a=1}^{\upsilon} \frac{1}{g_a^2}(EW + WE) + \vartheta + (1+m_2)EBB^T E - EH - HE \Big]$, *then the network (1) achieves the bipartite synchronization.*

Proof. Take a suitable Lyapunov functional for the system (3):

$$V(t) = \sum_{\epsilon=1}^{\zeta} \int_\Theta \varphi_\epsilon^T(\theta,t) E \varphi_\epsilon(\theta,t) d\theta. \tag{7}$$

On the basis of Lemma 2.1, we have

$$\begin{aligned} D^{\varpi} V(t) \leqslant{} & 2\sum_{\epsilon=1}^{\zeta} \int_\Theta \varphi_\epsilon^T(\theta,t) E \Big(W\Delta\varphi_\epsilon(\theta,t) - H\varphi_\epsilon(\theta,t) \\ & + s_\epsilon B\hbar(\kappa_\epsilon(\theta,t)) - B\hbar(\bar\kappa(\theta,t)) + B\hbar(\bar\kappa(\theta,t)) \\ & + s_\epsilon B\hbar(\kappa_\epsilon(\theta,t-\tau(t))) - B\hbar(\bar\kappa(\theta,t-\tau(t))) \\ & + B\hbar(\bar\kappa(\theta,t-\tau(t))) - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t)) \\ & + \sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta} d^v s_\epsilon s_\omega Q_{\epsilon\omega}^v P^v \varphi_\omega(\theta,t) \\ & - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t-\tau(t))) \Big) d\theta. \end{aligned} \tag{8}$$

Since

$$\begin{aligned} \sum_{\epsilon=1}^{\zeta} \varphi_\epsilon(\theta,t) &= \sum_{\epsilon=1}^{\zeta} \Big(s_\epsilon \kappa_\epsilon(\theta,t) - \frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q \kappa_q(\theta,t) \Big) \\ &= \sum_{\epsilon=1}^{\zeta} s_\epsilon \kappa_\epsilon(\theta,t) - \sum_{q=1}^{\zeta} s_q \kappa_q(\theta,t) \\ &= 0, \end{aligned}$$

one has

$$\sum_{\epsilon=1}^{\zeta} \varphi_\epsilon^T(\theta,t) B\hbar(\bar\kappa(\theta,t)) = 0, \tag{9}$$

$$\sum_{\epsilon=1}^{\zeta} \varphi_\epsilon^T(\theta,t) B\hbar(\bar\kappa(\theta,t-\tau(t))) = 0, \tag{10}$$

$$\sum_{\epsilon=1}^{\zeta} \varphi_\epsilon^T(\theta,t) \Big(\frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t)) \Big) = 0, \tag{11}$$

$$\sum_{\epsilon=1}^{\zeta} \varphi_\epsilon^T(\theta,t) \Big(\frac{1}{\zeta}\sum_{q=1}^{\zeta} s_q B\hbar(\kappa_q(\theta,t-\tau(t))) \Big) = 0. \tag{12}$$

From the boundary condition, Green's formula, Lemma 2.2, and (4), one acquires

$$\begin{aligned} & 2\int_\Theta \varphi_\epsilon^T(\theta,t) EW \Delta\varphi_\epsilon(\theta,t) d\theta \\ ={}& 2\sum_{c=1}^{\iota}\sum_{r=1}^{\iota} e_{cr} w_r \int_\Theta \varphi_{\epsilon c}(\theta,t) \Delta\varphi_{\epsilon r}(\theta,t) d\theta \\ ={}& -2\sum_{a=1}^{\upsilon}\sum_{c=1}^{\iota}\sum_{r=1}^{\iota} e_{cr} w_r \int_\Theta \frac{\partial\varphi_{\epsilon c}(\theta,t)}{\partial\theta_a} \frac{\partial\varphi_{\epsilon r}(\theta,t)}{\partial\theta_a} d\theta \\ ={}& -\sum_{a=1}^{\upsilon} \int_\Theta \Big(\frac{\partial\varphi_\epsilon(\theta,t)}{\partial\theta_a} \Big)^T (EW + WE) \frac{\partial\varphi_\epsilon(\theta,t)}{\partial\theta_a} d\theta \\ \leqslant{}& -\sum_{a=1}^{\upsilon} \frac{1}{g_a^2} \int_\Theta \varphi_\epsilon^T(\theta,t)(EW + WE)\varphi_\epsilon(\theta,t) d\theta. \end{aligned} \tag{13}$$

In view of (2), one derives

$$\begin{aligned} & 2\varphi_\epsilon^T(\theta,t) EB(s_\epsilon \hbar(\kappa_\epsilon(\theta,t)) - \hbar(\bar\kappa(\theta,t))) \\ \leqslant{}& \varphi_\epsilon^T(\theta,t) EBB^T E \varphi_\epsilon(\theta,t) \\ & + (s_\epsilon\hbar(\kappa_\epsilon(\theta,t)) - \hbar(\bar\kappa(\theta,t)))^T (s_\epsilon\hbar(\kappa_\epsilon(\theta,t)) - \hbar(\bar\kappa(\theta,t))) \\ \leqslant{}& \varphi_\epsilon^T(\theta,t) EBB^T E\varphi_\epsilon(\theta,t) + \varphi_\epsilon^T(\theta,t)\vartheta\varphi_\epsilon(\theta,t), \end{aligned} \tag{14}$$

$$\begin{aligned} & 2\varphi_\epsilon^T(\theta,t) EB(s_\epsilon\hbar(\kappa_\epsilon(\theta,t-\tau(t))) - \hbar(\bar\kappa(\theta,t-\tau(t)))) \\ \leqslant{}& m_2 \varphi_\epsilon^T(\theta,t) EBB^T E\varphi_\epsilon(\theta,t) + \frac{1}{m_2}(s_\epsilon\hbar(\kappa_\epsilon(\theta,t-\tau(t))) - \hbar(\bar\kappa(\theta,t-\tau(t))))^T \\ & \times (s_\epsilon\hbar(\kappa_\epsilon(\theta,t-\tau(t))) - \hbar(\bar\kappa(\theta,t-\tau(t)))) \\ \leqslant{}& m_2\varphi_\epsilon^T(\theta,t) EBB^T E\varphi_\epsilon(\theta,t) + \frac{1}{m_2}\varphi_\epsilon^T(\theta,t-\tau(t))\vartheta\varphi_\epsilon(\theta,t-\tau(t)). \end{aligned} \tag{15}$$

From (8)-(15), one can obtain

$$\begin{aligned} D^{\varpi}V(t) \leqslant{} & \sum_{\epsilon=1}^{\zeta} \int_\Theta \varphi_\epsilon^T(\theta,t) \Big[-\sum_{a=1}^{\upsilon} \frac{1}{g_a^2}(EW + WE) + \vartheta \\ & + (1+m_2)EBB^T E - EH - HE \Big] \varphi_\epsilon(\theta,t) d\theta \\ & + \sum_{v=1}^{\sigma} d^v \int_\Theta \varphi^T(\theta,t)[(SQ^v S)\otimes(EP^v + P^v E)]\varphi(\theta,t) d\theta \\ & + \frac{1}{m_2}\sum_{\epsilon=1}^{\zeta}\int_\Theta \varphi_\epsilon^T(\theta,t-\tau(t))\vartheta\varphi_\epsilon(\theta,t-\tau(t)) d\theta \\ ={} & \int_\Theta \varphi^T(\theta,t)\Big[\Xi + \sum_{v=1}^{\sigma} d^v (SQ^v S)\otimes(EP^v + P^v E)\Big]\varphi(\theta,t) d\theta \\ & + \frac{1}{m_2}\int_\Theta \varphi^T(\theta,t-\tau(t))(I_\zeta\otimes\vartheta)\varphi(\theta,t-\tau(t)) d\theta, \end{aligned}$$

in which $\varphi(\theta,t) = (\varphi_1^T(\theta,t), \varphi_2^T(\theta,t), \ldots, \varphi_\zeta^T(\theta,t))^T$.

Based on (5), it is easy to obtain $F = \frac{1}{m_2}\vartheta - m_2 E \leqslant 0$. Then, we can derive from (6) that

$$\begin{aligned} & D^{\varpi}V(t) + m_1 V(t) - m_2 \sup_{-\tau(t)\leqslant u\leqslant 0} V(t+u) \\ \leqslant{}& \int_\Theta \varphi^T(\theta,t)\Big[\Xi + m_1 I_\zeta\otimes E + \sum_{v=1}^{\sigma} d^v (SQ^v S)\otimes(EP^v + P^v E)\Big]\varphi(\theta,t) d\theta \\ & + \int_\Theta \varphi^T(\theta,t-\tau(t))(I_\zeta\otimes F)\varphi(\theta,t-\tau(t)) d\theta \\ \leqslant{}& 0. \end{aligned} \tag{16}$$

In light of Lemma 2.4 and (16), we have $\lim_{t\to+\infty} V(t) = 0$. Then, according to (7), we can derive

$$\lim_{t\to+\infty} ||\varphi_\epsilon(\cdot,t)|| = \lim_{t\to+\infty} \|s_\epsilon\kappa_\epsilon(\cdot,t) - \bar\kappa(\cdot,t)\| = 0,$$

in which $\epsilon = 1, 2, \ldots, \zeta$. Therefore, the network (1) can achieve the bipartite synchronization. □

Remark 1. In recent years, several authors have tackled the bipartite synchronization for CFRNN, and some significant results on this topic have been obtained (Hao et al., 2024; Wu et al., 2023a; Xu et al., 2024). Since various influencing factors, it is very meaningful to further discuss the bipartite synchronization for multiweighted CFRNN, but no result has been reported on this topic up to now. In this subsection, a bipartite synchronization criterion for the MCFDRNN (1) is derived by employing Lemma 2.4 (see Theorem 3.1).

3.3. $\mathcal{H}_\infty$ bipartite synchronization for MCFDRNN with external disturbances

Due to the existences of environmental factors and modeling errors, external disturbances commonly exist in MCFDRNN and may destroy the bipartite synchronization. Apparently, it is essential to further discuss the external disturbances when investigating the MCFDRNN. Then, the external disturbances are incorporated into the MCFDRNN (1), which can be expressed as follows:

$$\begin{aligned}D^{\varpi}\kappa_\epsilon(\theta,t)&=W\Delta\kappa_\epsilon(\theta,t)-H\kappa_\epsilon(\theta,t)+B\hbar(\kappa_\epsilon(\theta,t))\\&\quad+\sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta}d^vQ^v_{\epsilon\omega}P^v\kappa_\omega(\theta,t)+\mu_\epsilon(\theta,t)\\&\quad+B\hbar(\kappa_\epsilon(\theta,t-\tau(t))),\ \epsilon=1,2,\dots,\zeta,\\\kappa_\epsilon(\theta,t)&=0,(\theta,t)\in\partial\Theta\times[-\psi,+\infty),\\\kappa_\epsilon(\theta,t)&=\tilde{\Psi}_\epsilon(\theta,t),(\theta,t)\in\Theta\times[-\psi,0],\end{aligned}\tag{17}$$

in which $\tilde{\Psi}_\epsilon(\theta,t)$ is continuous on $\bar{\Theta}\times[-\psi,0]$, and $\mu_\epsilon(\theta,t)\in\mathbb{R}^\iota$ denotes the external disturbance and fulfills $I^{\varpi}\|\mu_\epsilon(\cdot,t)\|^2<+\infty$ for any $0\leqslant t\in\mathbb{R}$.

Remark 2. Recently, CRNN with multiple weights has attracted much attention of researchers since the diversity of influencing factors, and plenty of significant results about the dynamical behaviors of multi-weighted CRNN have been obtained Cao et al. (2022), Lin and Liu (2023), Zhao et al. (2023). Considering that fractional-order derivative can better model the dynamics of CRNN compared with integer-order derivative and external disturbances commonly exist in CRNN due to the influences of environmental factors and modeling errors, a multi-weighted CFDRNN with external disturbances is introduced in this subsection.

Take $\varphi_\epsilon(\theta,t)=s_\epsilon\kappa_\epsilon(\theta,t)-\frac{1}{\zeta}\sum_{q=1}^{\zeta}s_q\kappa_q(\theta,t),\ \epsilon=1,2,\dots,\zeta$. In light of (17), we can derive

$$\begin{aligned}D^{\varpi}\varphi_\epsilon(\theta,t)&=W\Delta\varphi_\epsilon(\theta,t)-H\varphi_\epsilon(\theta,t)+s_\epsilon B\hbar(\kappa_\epsilon(\theta,t))\\&\quad+s_\epsilon B\hbar(\kappa_\epsilon(\theta,t-\tau(t)))-\frac{1}{\zeta}\sum_{q=1}^{\zeta}s_qB\hbar(\kappa_q(\theta,t))\\&\quad+\sum_{v=1}^{\sigma}\sum_{\omega=1}^{\zeta}d^vs_\epsilon s_\omega Q^v_{\epsilon\omega}P^v\varphi_\omega(\theta,t)\\&\quad+s_\epsilon\mu_\epsilon(\theta,t)-\frac{1}{\zeta}\sum_{q=1}^{\zeta}s_q\mu_q(\theta,t)\\&\quad-\frac{1}{\zeta}\sum_{q=1}^{\zeta}s_qB\hbar(\kappa_q(\theta,t-\tau(t))),\end{aligned}\tag{18}$$

in which $\epsilon=1,2,\dots,\zeta$.

In what follows, a definition about the $\mathcal{H}_\infty$ bipartite synchronization for the network (17) is introduced.

Definition 3. The network (17) is $\mathcal{H}_\infty$ bipartite synchronized if

$$\sum_{\epsilon=1}^{\zeta}\|\varphi_\epsilon(\cdot,t)\|^2\leqslant-D^{\varpi}\Phi(t)+h\sup_{-\tau(t)\leqslant u\leqslant0}\Phi(t+u)+\chi^2\sum_{\epsilon=1}^{\zeta}\|\mu_\epsilon(\cdot,t)\|^2\tag{19}$$

for any $0\leqslant t\in\mathbb{R}$, in which $0\leqslant\Phi(\cdot)\in\mathbb{R}$, $0<h\in\mathbb{R}$, and $0<\chi\in\mathbb{R}$.

Remark 3. Because external disturbances are inevitable in MCFDRNN and may destroy the bipartite synchronization, it is essential to further discuss the $\mathcal{H}_\infty$ bipartite synchronization for the MCFDRNN (17). In fact, some authors have preliminary addressed the $\mathcal{H}_\infty$ synchronization in CFRNN (Wang et al., 2025). On the basis of these existing results (Wang et al., 2025), a novel $\mathcal{H}_\infty$ bipartite synchronization definition for the network (17) is introduced (see Definition 3.2).

The following theorem provides a criterion for the network (17) to achieve the $\mathcal{H}_\infty$ bipartite synchronization.

Theorem 2. *If there exist $0<\chi\in\mathbb{R}$, $0<m_2\in\mathbb{R}$, and $0<E=(e_{cr})_{\iota\times\iota}\in\mathbb{R}^{\iota\times\iota}$ satisfying*

$$EW+WE\geqslant0,\tag{20}$$

$$m_2\geqslant\sqrt{\frac{\xi_1(\vartheta)}{\xi_\iota(E)}},\tag{21}$$

$$\bar{\Xi}+I_{\zeta\iota}+\frac{1}{\chi^2}S^2\otimes E^2\leqslant0,\tag{22}$$

in which $\bar{\Xi}=I_\zeta\otimes\Big[-\sum_{a=1}^{\upsilon}\frac{1}{g_a^2}(EW+WE)+\vartheta+(1+m_2)EBB^TE-EH-HE\Big]+\sum_{v=1}^{\sigma}d^v(SQ^vS)\otimes(EP^v+P^vE)$*, then the network (17) is $\mathcal{H}_\infty$ bipartite synchronized.*

Proof. Take the Lyapunov functional as

$$V(t)=\sum_{\epsilon=1}^{\zeta}\int_\Theta\varphi_\epsilon^T(\theta,t)E\varphi_\epsilon(\theta,t)d\theta.\tag{23}$$

On the basis of Lemma 2.1, one acquires

$$\begin{aligned}D^{\varpi}V(t)&\leqslant\int_\Theta\varphi^T(\theta,t)\bar{\Xi}\varphi(\theta,t)d\theta\\&\quad+\frac{1}{m_2}\int_\Theta\varphi^T(\theta,t-\tau(t))(I_\zeta\otimes\vartheta)\varphi(\theta,t-\tau(t))d\theta\\&\quad+2\int_\Theta\varphi^T(\theta,t)(S\otimes E)\mu(\theta,t)d\theta.\end{aligned}\tag{24}$$

In view of (22), (21), and (24), one has

$$\begin{aligned}&D^{\varpi}V(t)+\sum_{\epsilon=1}^{\zeta}\|\varphi_\epsilon(\cdot,t)\|^2-\chi^2\sum_{\epsilon=1}^{\zeta}\|\mu_\epsilon(\cdot,t)\|^2-m_2\sup_{-\tau(t)\leqslant u\leqslant0}V(t+u)\\&\leqslant\int_\Theta\varphi^T(\theta,t)(\bar{\Xi}+I_{\zeta\iota})\varphi(\theta,t)d\theta\\&\quad+\int_\Theta\varphi^T(\theta,t-\tau(t))(I_\zeta\otimes F)\varphi(\theta,t-\tau(t))d\theta\\&\quad+2\int_\Theta\varphi^T(\theta,t)(S\otimes E)\mu(\theta,t)d\theta-\chi^2\int_\Theta\mu^T(\theta,t)\mu(\theta,t)d\theta\\&\leqslant\int_\Theta\varphi^T(\theta,t)\Big(\bar{\Xi}+I_{\zeta\iota}+\frac{1}{\chi^2}S^2\otimes E^2\Big)\varphi(\theta,t)d\theta\\&\quad+\int_\Theta\varphi^T(\theta,t-\tau(t))(I_\zeta\otimes F)\varphi(\theta,t-\tau(t))d\theta\\&\quad-\Big[\frac{1}{\chi}(S\otimes E)\varphi(\theta,t)-\chi\mu(\theta,t)\Big]^T\Big[\frac{1}{\chi}(S\otimes E)\varphi(\theta,t)-\chi\mu(\theta,t)\Big]d\theta\\&\leqslant\int_\Theta\varphi^T(\theta,t)\Big(\bar{\Xi}+I_{\zeta\iota}+\frac{1}{\chi^2}S^2\otimes E^2\Big)\varphi(\theta,t)d\theta\\&\quad+\int_\Theta\varphi^T(\theta,t-\tau(t))(I_\zeta\otimes F)\varphi(\theta,t-\tau(t))d\theta\\&\leqslant0,\end{aligned}\tag{25}$$

in which $\mu(\theta,t)=(\mu_1^T(\theta,t),\mu_2^T(\theta,t),\dots,\mu_\zeta^T(\theta,t))^T$ and $F=\frac{1}{m_2}\vartheta-m_2E$.

From (25), one acquires

$$\sum_{\epsilon=1}^{\zeta}\|\varphi_\epsilon(\cdot,t)\|^2\leqslant-D^{\varpi}V(t)+m_2\sup_{-\tau(t)\leqslant u\leqslant0}V(t+u)+\chi^2\sum_{\epsilon=1}^{\zeta}\|\mu_\epsilon(\cdot,t)\|^2.$$

Therefore, the network (17) is $\mathcal{H}_\infty$ bipartite synchronized. □

Remark 4. Considering that external disturbances widely exist in MCFDRNN, it is essential and meaningful to further investigate the $\mathcal{H}_\infty$ bipartite synchronization for MCFDRNN with external disturbances. But, no results have been reported on this topic up to now. In this subsection, an $\mathcal{H}_\infty$ bipartite synchronization criterion for the MCFDRNN (17) is derived by employing the properties of Caputo fractional-order derivative and matrix theory (see Theorem 3.2).

4. Numerical example

A numerical example is carried out to verify the correctness of the derived bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization criteria in this section.

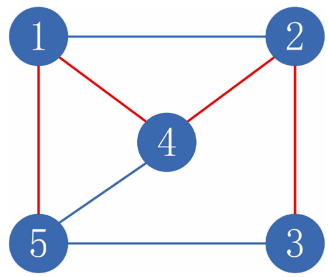

Fig. 1. The topology of the MCFDRNN (26).

Example 4.1: A MCFDRNN consisting of 5 fractional-order delayed reaction-diffusion neural networks is characterized as follows:

$$\begin{aligned} D^{\varpi}\kappa_\epsilon(\theta,t) &= W\frac{\partial^2\kappa_\epsilon(\theta,t)}{\partial\theta^2} - H\kappa_\epsilon(\theta,t) + B\hbar(\kappa_\epsilon(\theta,t)) \\ &+ \sum_{\nu=1}^{3}\sum_{\omega=1}^{5} d^\nu Q^\nu_{\epsilon\omega}P^\nu\kappa_\omega(\theta,t) \\ &+ B\hbar(\kappa_\epsilon(\theta,t-\tau(t))),\ \epsilon = 1,2,\dots,5, \end{aligned} \tag{26}$$

in which $\theta\in(-0.5,0.5)$, $d^1=8$, $d^2=6$, $d^3=7$, $W=\mathrm{diag}(0.8,0.6,0.5)$, $H=\mathrm{diag}(0.4,0.3,0.6)$, $\hbar_c(x)=0.4(|x+1|-|x-1|)$, $x\in\mathbb{R}, c=1,2,3$, $P^1=\mathrm{diag}(0.6,0.5,0.7)$, $P^2=\mathrm{diag}(0.8,0.6,0.5)$, $P^3=\mathrm{diag}(0.9,0.3,0.2)$, $\tau(t)=0.2-0.2e^{-0.4t}$,

$$B=\begin{pmatrix} 0.5 & 0.7 & -0.3 \\ 0.4 & 0.2 & 0.8 \\ 0.6 & 0.3 & -0.9 \end{pmatrix},$$

$$Q^1=\begin{pmatrix} -1.1 & 0.4 & 0 & -0.3 & -0.4 \\ 0.4 & -1.2 & -0.3 & -0.5 & 0 \\ 0 & -0.3 & -0.5 & 0 & 0.2 \\ -0.3 & -0.5 & 0 & -1.1 & 0.3 \\ -0.4 & 0 & 0.2 & 0.3 & -0.9 \end{pmatrix},$$

$$Q^2=\begin{pmatrix} -0.9 & 0.2 & 0 & -0.2 & -0.5 \\ 0.2 & -0.9 & -0.3 & -0.4 & 0 \\ 0 & -0.3 & -0.5 & 0 & 0.2 \\ -0.2 & -0.4 & 0 & -0.9 & 0.3 \\ -0.5 & 0 & 0.2 & 0.3 & -1 \end{pmatrix},$$

$$Q^3=\begin{pmatrix} -0.8 & 0.1 & 0 & -0.4 & -0.3 \\ 0.1 & -0.8 & -0.2 & -0.5 & 0 \\ 0 & -0.2 & -0.7 & 0 & 0.5 \\ -0.4 & -0.5 & 0 & -1.2 & 0.3 \\ -0.3 & 0 & 0.5 & 0.3 & -1.1 \end{pmatrix}.$$

Apparently, $\hbar_c(\cdot)$, $c=1,2,3$ satisfy (2) with $\vartheta_c=0.8$. The topology of MCFDRNN (26) is shown in Fig. 1, which is apparently connected. From Fig. 1, $\mathcal{V}=\{1,2,3,4,5\}$ can be divided into the subsets $\mathcal{V}_1=\{1,2\}$ and $\mathcal{V}_2=\{3,4,5\}$ with $\mathcal{V}_1\cap\mathcal{V}_2=\varnothing$.

Case 1: By exploiting the MATLAB YALMIP Toolbox, one obtains $m_1=1.2165$, $m_2=1.0612$, and

$$E=\begin{pmatrix} 0.6250 & -0.0000 & -0.0000 \\ -0.0000 & 0.8333 & -0.0000 \\ -0.0000 & -0.0000 & 1.0000 \end{pmatrix}$$

satisfying (4)-(6). From Theorem 3.1, the MCFDRNN (1) is bipartite synchronized. The evolutions of $\|\varphi_\epsilon(\cdot,t)\|, \epsilon=1,2,\dots,5$ are depicted in Fig. 2 and the trajectories of $\kappa_{\epsilon\iota}(\cdot,t), \epsilon=1,2,\dots,5, \iota=1,2,3$ are depicted in Figs. 3,4,5.

Case 2: Take $\mu_\epsilon(\theta,t)=(\epsilon|\sin(\frac{2\pi}{3}t)|\cos(\theta\pi), 2\epsilon|\sin(\frac{2\pi}{3}t)|\cos^2(\theta\pi), 4\sqrt{\epsilon}|\sin(\frac{2\pi}{3}t)|\cos^2(\theta\pi))^T$, $\epsilon=1,2,\dots,5$, By employing the MATLAB YALMIP Toolbox, one gets $\chi=2.4421$, $m_2=1.0604$, and

$$E=\begin{pmatrix} 0.6250 & -0.0000 & -0.0000 \\ -0.0000 & 0.8333 & -0.0000 \\ -0.0000 & -0.0000 & 1.0000 \end{pmatrix}$$

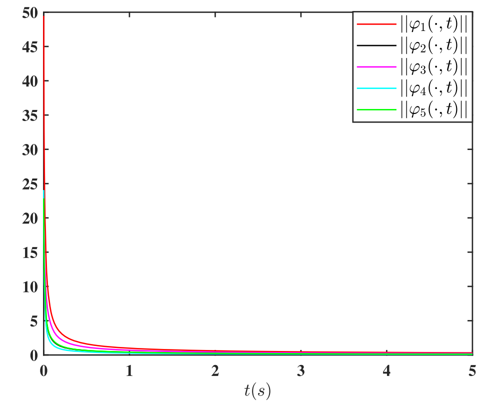

Fig. 2. The change processes of $\|\varphi_\epsilon(\cdot,t)\|, \epsilon=1,2,\dots,5$.

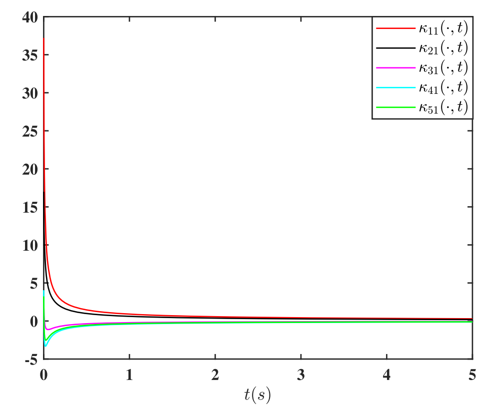

Fig. 3. The trajectories of $\kappa_{\epsilon1}(\cdot,t), \epsilon=1,2,\dots,5$.

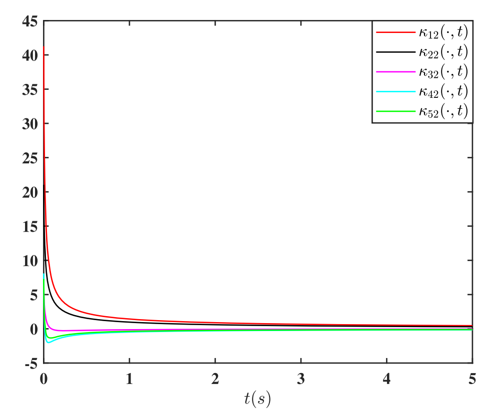

Fig. 4. The trajectories of $\kappa_{\epsilon2}(\cdot,t), \epsilon=1,2,\dots,5$.

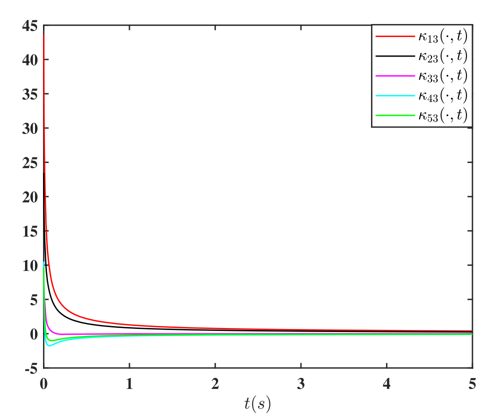

Fig. 5. The trajectories of $\kappa_{\epsilon3}(\cdot,t), \epsilon=1,2,\dots,5$.

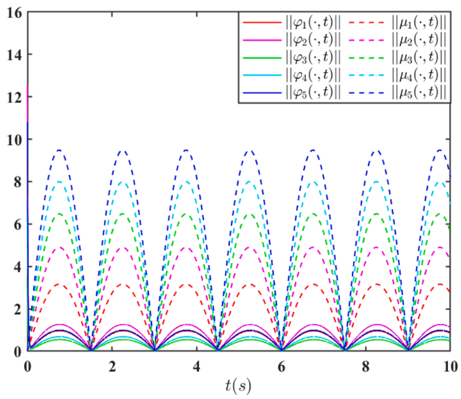

Fig. 6. $\|\varphi_\epsilon(\cdot,t)\|$ and $\|\mu_\epsilon(\cdot,t)\|$, $\epsilon = 1,2,\ldots,5$.

fulfilling (20)-(22). From Theorem 3.2, the MCFDRNN (26) with external disturbances is $\mathcal{H}_\infty$ bipartite synchronized. The change processes of $\|\varphi_\epsilon(\cdot,t)\|$ and $\|\mu_\epsilon(\cdot,t)\|$, $\epsilon = 1,2,\ldots,5$ are portrayed in Fig. 6.

Remark 5. Practically, the bipartite synchronization phenomenon widely exists in numerous real networks, such as mobile multi-robot networks Wu et al. (2023a). From Fig. 3, we can clearly see that the state variables of nodes 1 and 2 (3, 4, and 5) converge to the same values, but the state variables of nodes in different subsets {1,2} and {3,4,5} tend to opposite values. For the convenience of readers, a mobile multi-robot network consisting of five robots is given to explain Fig. 3. Five robots are divided into two groups {1,2} and {3,4,5}, Fig. 3 means that robots 1 and 2 (3,4, and 5) move towards the same direction at the same speed, but robots in two different groups {1,2} and {3,4,5} move towards the opposite direction at the same speed.

5. Conclusion

The bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization for the MCFDRNN have been addressed in this paper. By employing the properties of Caputo fractional-order derivative and inequality techniques, several bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization criteria for the MCFDRNN without and with external disturbances have been derived. Finally, the derived bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization criteria have been validated by a numerical example. In the future, we shall tackle the analysis and impulsive control for the bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization of coupled fractional-order delayed reaction-diffusion neural networks with multiple output or spatial diffusion couplings.

In this paper, a bipartite synchronization and an $\mathcal{H}_\infty$ bipartite synchronization criteria in the form of matrix inequalities have been obtained, in which the dimensions of these matrix inequalities depend on the number of nodes and the dimension of node state. Apparently, when the number of nodes is very large, the dimensions of matrix inequalities may be very high such that they are very difficult to solve. In the future, we shall try our best to adopt other methods and techniques to acquire several bipartite synchronization and $\mathcal{H}_\infty$ bipartite synchronization criteria which not only have lower dimensions but also are easier to solve.

CRediT authorship contribution statement

Ya-Nan Li: Writing – original draft; **Jin-Liang Wang:** Writing – original draft; **Shun-Yan Ren:** Writing – original draft; **Tingwen Huang:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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